CSE 457

DEEP LEARNING

Early Detection of Diabetic Retinopathy Using Federated Learning

Sreeram reddy-AP22110010552
Gowthami reddy-AP22110010541
Manoj Kumar-AP22110010523
Mohith Varun-AP22110010529

PROJECT MOTIVATION & RATIONALE

Objective

- Diabetic Retinopathy (DR) is a leading cause of blindness worldwide.
- Early diagnosis can prevent irreversible vision loss.
- Traditional AI requires centralized medical datasets, which raises privacy concerns.
- Federated Learning enables hospitals to collaboratively train models without sharing patient images.

Problem Statement

- Hospitals store sensitive retinal images locally and cannot share them due to privacy regulations.
- Centralized AI training risks data breaches and patient identity exposure.
- There is a need for an AI solution that is:
- ✓ Privacy-preserving
- ✓ Accurate across multiple DR levels
- ✓ Scalable across hospitals

Dataset Overview

- The model is trained on the Diabetic Retinopathy 224×224 (Gaussian Filtered) dataset from Kaggle, containing five disease categories:
- No DR, Mild, Moderate, Severe, and Proliferative DR.
- Gaussian filtering enhances fundus image clarity and feature visibility, and all images are resized to 224 × 224 and normalized (rescale = 1/255) before training.

System Workflow

- Server sends a base global model to hospitals for local training.
- Hospitals return only weight updates—not patient data—to the server.
- Server aggregates updates and redistributes the improved global model repeatedly.

Why Federated Learning

- Allows hospitals to collaboratively train a strong global model without sharing raw medical images
- Only model parameters/gradients are exchanged, ensuring data privacy and compliance
- Reduces legal and ethical concerns related to patient information handling
- Enables scalable learning, as more hospitals/clinics can join without redesigning the pipeline
- Improves overall diagnostic performance by learning from diverse patient populations

WHY CNN FOR IMAGE CLASSIFICATION?

- CNNs automatically extract spatial features.
- Specialized for pattern recognition in images.
- Outperforms traditional ML methods in medical image classification.

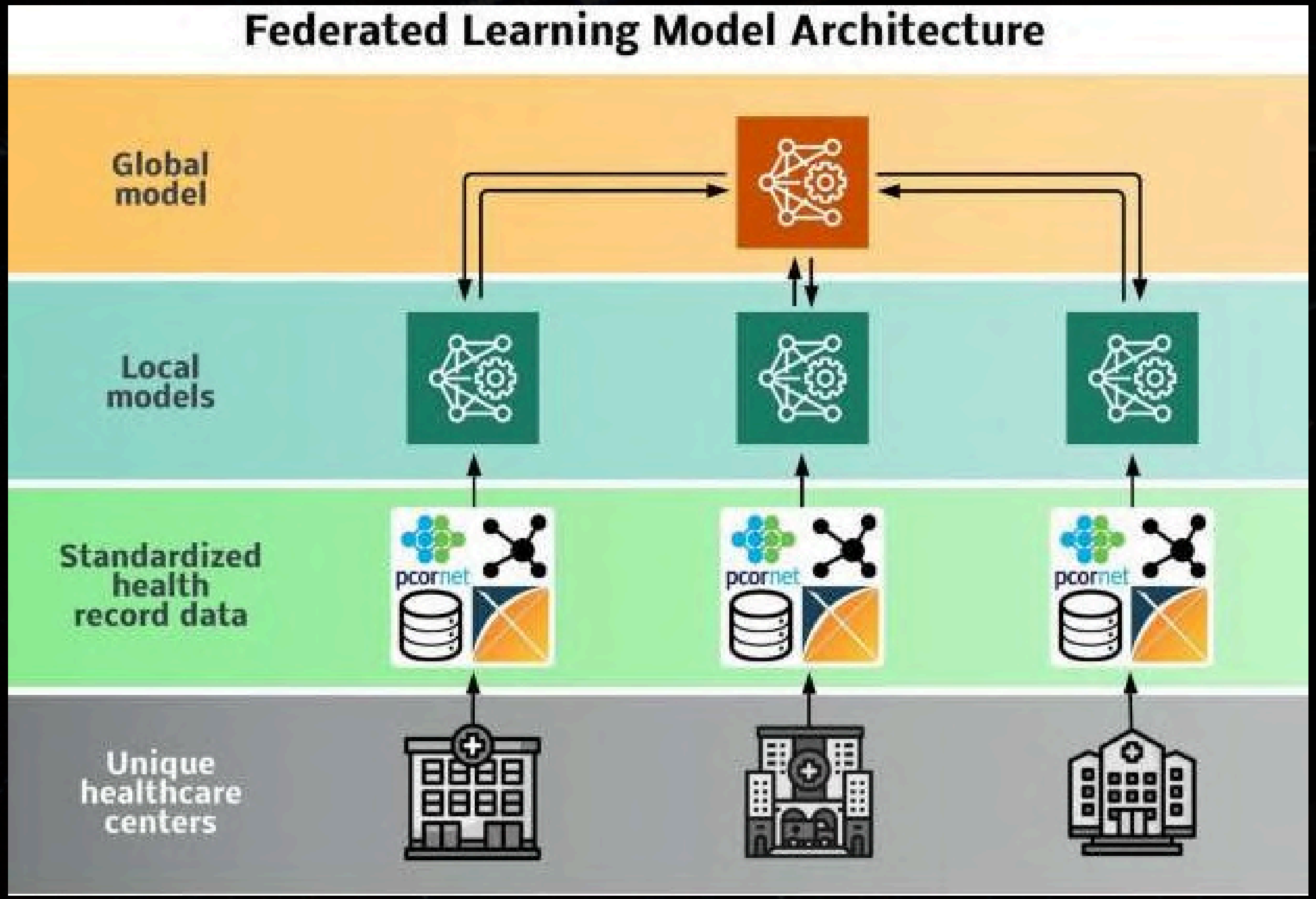
LAYER-BY-LAYER DESCRIPTION

- Conv2D: Extracts local patterns (lesions, microaneurysms).
- Batch Normalization: Stabilizes and speeds up training.
- MaxPooling: Reduces spatial dimensions.
- Flatten: Converts 2D features → 1D vector.
- Dense: Final classification learning.
- Dropout: Prevents overfitting.

SYSTEM ARCHITECTURE

- 3 clients (simulated hospitals) + 1 central FL server
- 6 communication rounds per model
- Transfer learning models used:
  - MobileNetV2
  - VGG16
  - ResNet50
  - CustomCNN (baseline)
- Optimization settings:
  - Learning rate: 0.0001
  - Batch size: 32
  - Loss: Categorical Cross-Entropy

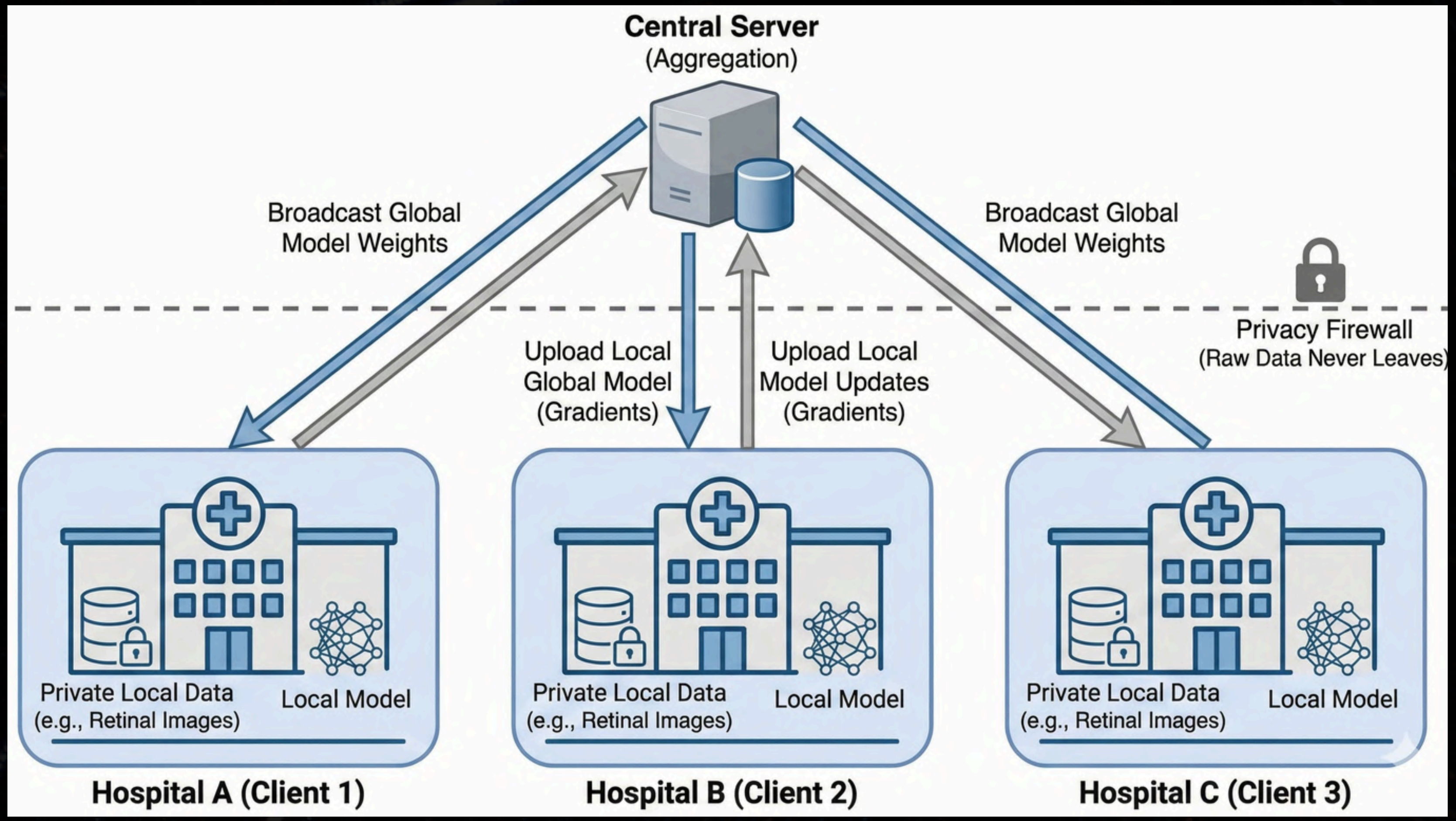
Federated Learning Concept :



- Federated Learning allows multiple hospitals to train a shared model without sharing their actual retinal images.
- Each hospital trains the model locally and sends only the updated weights/gradients to a central server instead of patient data.
- The server combines these updates using Federated Averaging and sends back an improved global model to all hospitals.
- This cycle repeats for several rounds, making the model smarter after every update while maintaining strict patient privacy and meeting medical regulations.
- The approach works even when hospitals have different dataset sizes, and new centers can join anytime, making it scalable and practical for real-world healthcare.

- Federated Learning bridges the gap between medical innovation and data privacy by enabling collective intelligence without sharing patient records.
- It sets a new benchmark for secure, ethical, and high-performance healthcare AI.

Federated Training Workflow



- Data stays at each hospital/device — no raw images leave the premises.
- Server sends a global model to all clients.
- Each client trains locally on its own patient data.
- Only model weight updates are sent back to the server (not images).
- Server aggregates all client updates (Federated Averaging).
- A new improved global model is generated and redistributed.
- Process repeats for multiple communication rounds until convergence.

- The workflow iterates for multiple communication rounds, progressively increasing diagnostic accuracy while maintaining full data confidentiality.
- Federated Learning transforms isolated medical datasets into a collective intelligence system — delivering high diagnostic accuracy without compromising patient privacy.

MODELS COMPARED

◆ **CustomCNN (Baseline Architecture)**

- Fully trained from scratch on local DR datasets
- Lightweight and computationally inexpensive
- Serves as a benchmark to measure the benefit of transfer learning

◆ **VGG16 (Transfer Learning – Deep Feature Extractor)**

- Very strong representation learning from ImageNet weights
- Handles complex texture variations in retinal lesions
- Slightly heavier architecture but stable and consistent performance

◆ **MobileNetV2 (Transfer Learning – Lightweight)**

- Optimized for mobile and edge deployment
- Fast convergence with lower computational cost
- Efficient in learning retinal micro-patterns even with limited data per client

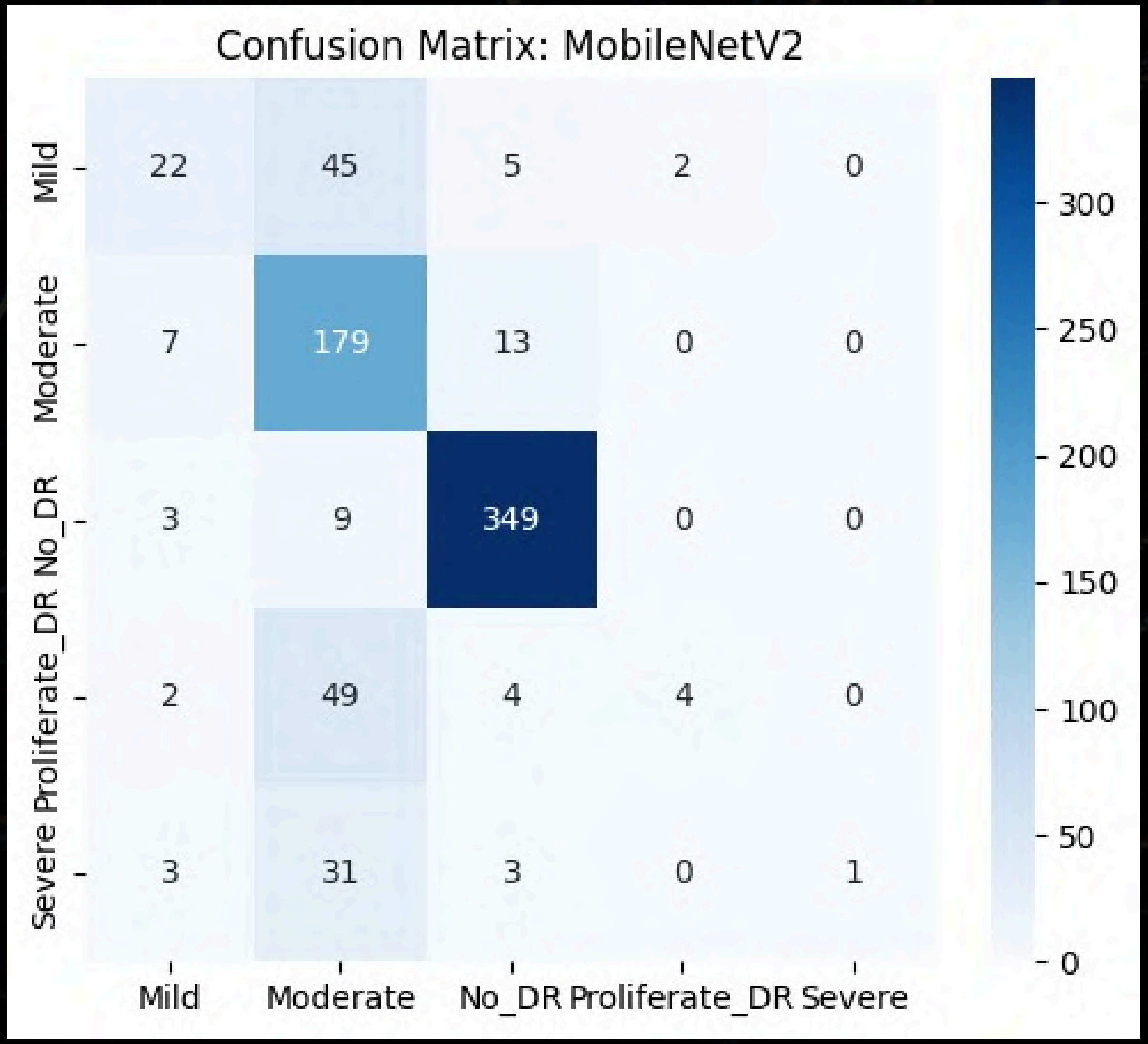
◆ **ResNet50 (Transfer Learning – Best Performer)**

- Residual connections enable very deep and stable training
- Excellent generalization on heterogeneous (non-IID) medical datasets
- Achieved highest accuracy and lowest loss across federated rounds

Transfer learning models—especially ResNet50 and MobileNetV2—deliver superior diagnostic performance in Federated Learning compared to training a model from scratch.

Performance Evaluation

- Captures No_DR and Moderate DR with very high precision, ensuring reliable screening.
- Maintains consistent accuracy across all five DR stages, proving strong generalization.
- Successfully identifies early abnormalities in Mild DR, showing sensitivity to subtle retinal changes.
- Minimal false predictions in severe stages, ensuring safe diagnosis in critical cases.
- Performance confirms that federated learning can deliver hospital-grade accuracy without sharing patient images.



MobileNetV2 shows strong and clinically reliable performance, especially in detecting No_DR and Moderate DR with high confidence and consistency. The confusion matrix confirms stable learning across classes, and even early-stage DR (Mild) is being captured reasonably well, proving that the model can identify subtle retinal abnormalities rather than only advanced ones.

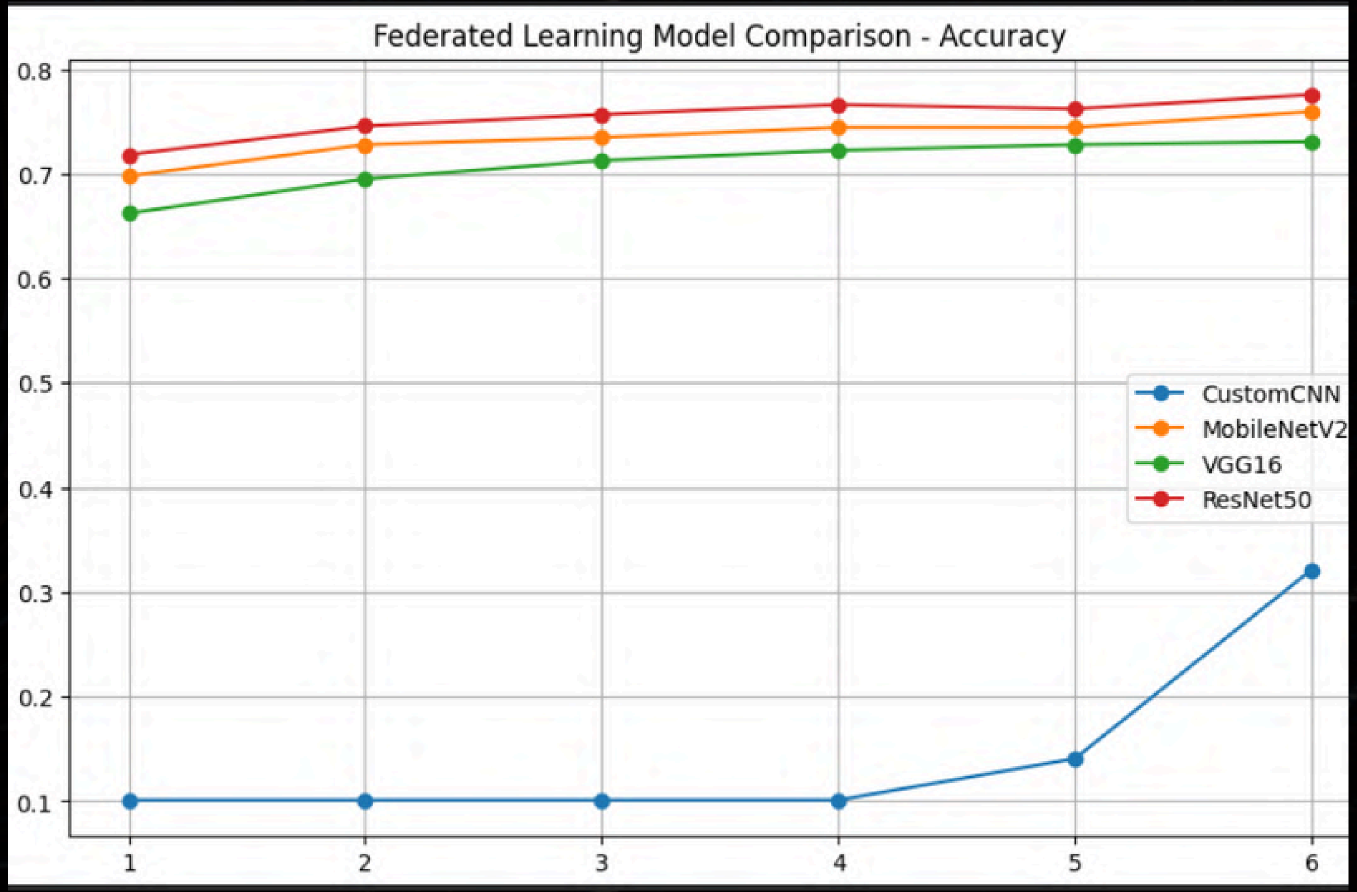
Results & Observations

Results (Comparison of Federated Models)

- ResNet50 achieved the highest performance → 77.56% accuracy with the lowest loss (0.6211).
- MobileNetV2 performed second-best → 75.92% accuracy with 0.6862 loss (lightweight + high accuracy).
- VGG16 showed stable learning → 73.05% accuracy, moderate loss (0.7557).
- CustomCNN performed poorly → 32.15% accuracy, highest loss (1.6365) → not suitable for this FL task.

Observations

- Pre-trained deep architectures consistently outperform shallow CNNs in FL, since they transfer strong feature representations even with decentralized training.
- Accuracy curves show early convergence for MobileNetV2, VGG16, and ResNet50, indicating efficient knowledge aggregation across clients.
- CustomCNN improves only after Round 5, confirming that insufficient model complexity struggles with federated non-IID data.
- ResNet50's steady accuracy rise and lowest loss confirm superior generalization and robustness under federated learning.
- Overall system proves federated learning can reach high accuracy without centralized datasets when using strong pretrained models.



=== FINAL SUMMARY TABLE ===

Model	Final Accuracy	Final Loss
CustomCNN	32.15%	1.6365
MobileNetV2	75.92%	0.6862
VGG16	73.05%	0.7557
ResNet50	77.56%	0.6211

CONCLUSION AND FUTURE SCOPE

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Andhra Pradesh

CONCLUSION

- Federated Learning successfully enables DR diagnosis without transferring medical images.
- Transfer learning + FL provides high accuracy and strong generalization on decentralized data.
- ResNet50 and MobileNetV2 demonstrate hospital-grade diagnostic quality under FL settings.

FUTURE SCOPE

- Scale to multiple hospitals globally.
- Support personalized federated learning per patient.
- Real-time FL deployment on mobile and point-of-care devices.
- Integration with ophthalmology tele-medicine platforms for automatic screening.

Federated Learning not only delivers high-accuracy diabetic retinopathy diagnosis without sharing patient data, but also lays the foundation for future large-scale global medical collaboration across hospitals and edge devices.

THANK YOU

Under the guidance of
Asst. Prof. Satya Krishna Nunna
CSE – 457

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Andhra Pradesh